From Table 19.2 it can be seen that the highest η value is exhibited by ^{233}U for thermal neutron energies and by ^{239}Pu for fast neutron energies. This suggests two different types of breeder reactor, the *thermal breeder* based on the reaction $^{232}\text{Th} \rightarrow {}^{233}\text{U}$ and the *fast breeder* based on the reaction $^{238}\text{U} \rightarrow {}^{239}\text{Pu}$. Due to the smaller effective cross sections, these breeders must be charged with a fuel having high concentrations (15−30%) of ^{233}U or Pu. These materials can be produced in conventional non-breeder reactors. Because a single inelastic scattering in ^{235}U may reduce neutron energy considerably, a fast breeder should only contain little of this isotope and hence depleted uranium is preferred over natural uranium as fertile material.

For a large fast breeder reactor like the French Creys-Malville (in operation from 1986 to 1997), the fuel utilization is higher than in the LWRs; a 1% burnup corresponds to about 8500 $\text{MW}_{\text{th}}\text{d t}^{-1}$ IHM. Burnup figures as high as 15% (130 000 $\text{MW}_{\text{th}}\text{d t}^{-1}$ IHM) have been achieved in fast breeders. Breeder reactors are described in Chapter 20.

19.10. THE OKLO PHENOMENON

At the Oklo mine in Gabon uranium has been mined for many years and delivered to the Pierrelatte isotope enrichment plant in France. The ore body, which is estimated to contain 400 000 t of uranium, is quite inhomogeneous, and pockets very rich in uranium are found embedded in sandstone and granite, see Figure 19.10. These pockets are often shaped like a lens, 10 m long and about 1 m in diameter, and contain on the average 10−20% pitchblende, although uranium concentrations up to 85% (pure pitchblende) are found in some spots. In 1972 it was discovered that the isotopic composition of some of the uranium received in France deviated from uranium from other sources. The ^{235}U content was significantly lower than the natural 0.72%. Careful analysis showed that some deliveries contained <0.5% ^{235}U, a serious disadvantage in materials to be used for ^{235}U enrichment. Analysis at Oklo showed samples even lower in ^{235}U content, as well as other elements whose isotopic composition considerably deviated from the natural one. For example, natural neodymium contains 27% ^{142}Nd, while Oklo neodymium contained <2%, see Fig. 19.10 and 19.11. On the other hand, natural neodymium contains 12% ^{143}Nd, while in Oklo, samples containing up to 24% ^{143}Nd were found. Fission product neodymium contains about 29% ^{143}Nd, while ^{142}Nd is not produced by fission. This should be compared with the lack of ^{142}Nd and the excess of ^{143}Nd in the Oklo samples, a condition which was found to be directly related to a high total concentration of uranium but a deficiency of ^{235}U. The conclusion was obvious: in some ancient time the missing ^{235}U had fissioned, producing ^{143}Nd among other fission products. This conclusion was supported by similar investigations on the isotopic composition of other elements produced by fission.

Because ^{235}U has a shorter half-life than ^{238}U, all uranium ores were richer in ^{235}U in the past. From $^{87}\text{Rb}-{}^{87}\text{Sr}$ analysis the age of the Oklo deposit is known to be 1.74×10^9 y; at

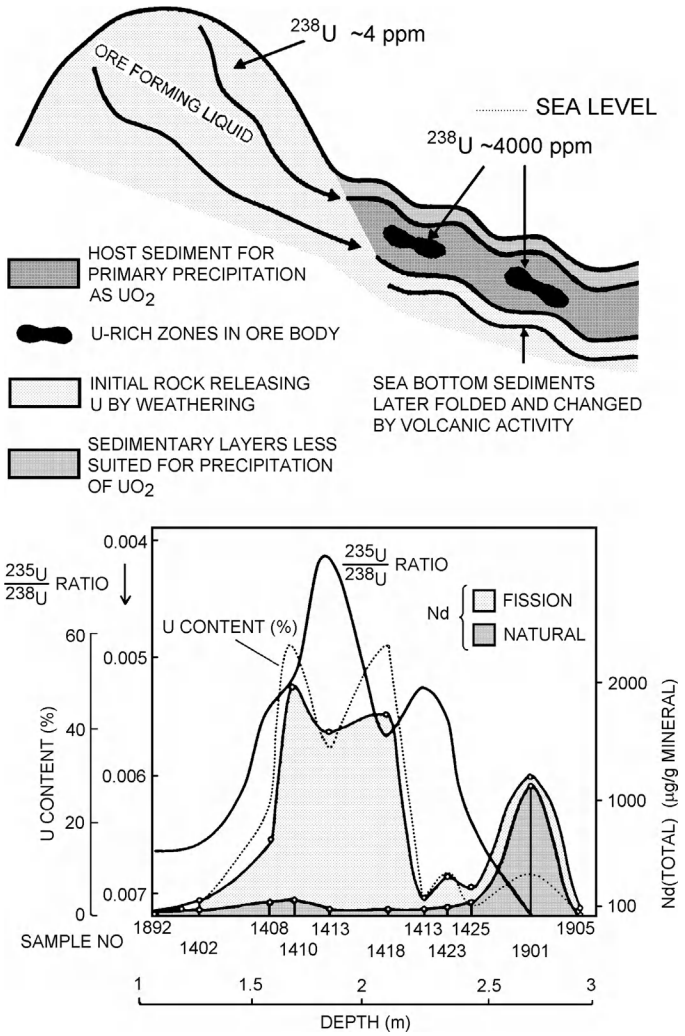


Figure 19.10 Schematic view indicating the formation of the Oklo ore deposit and some data on its current local composition.

that time the ^{235}U content of natural uranium was 3%. Although the fission factor η rapidly increases with ^{235}U content (about 1.8 for 3% ^{235}U), conditions in the natural Oklo deposit were such ($\varepsilon \sim 1.0$, $p \sim 0.4$, $f \sim 1.0$) that $k_\infty < 1$. The deposit is sedimentary and was formed in the presence of water, which greatly increases the resonance escape probability factor p; for an atomic ratio H_2O:U of 3:1, $p \sim 0.8$, and $k_\infty > 1$. Thus conditions existed in the past for a spontaneous, continuing chain reaction to occur in the Oklo deposit.

Further analysis and calculations have shown that these natural Oklo reactors (similar conditions occurred at several places) lasted for $\leq 10^6$ y. Probably, criticality occurred

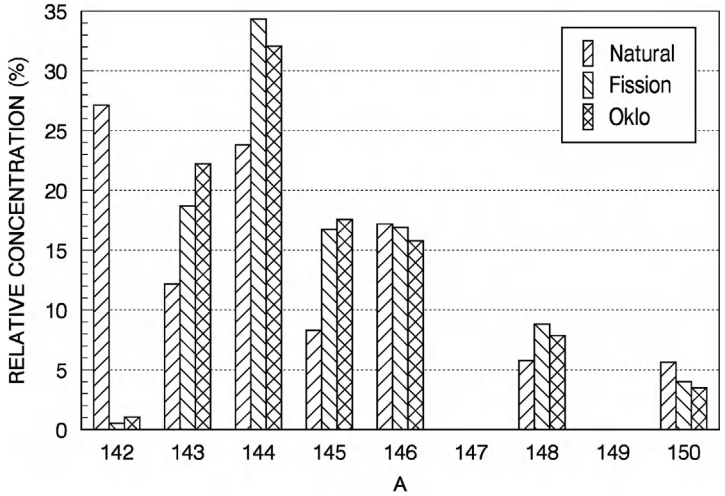


Figure 19.11 Comparison of the isotopic composition of neodymium from normal Nd ore, from ^{235}U fission in a BWR and from Oklo.

periodically as the heat from fission boiled away the water so that the chain reaction ceased after a while. When water returned (as the temperature decreased) the chain reaction would resume. The neutron flux probably never exceeded 10^{13} n m^{-2} s^{-1}.[5] The fluence is estimated to exceed 1.5×10^{25} n m^{-2}. This would have consumed about 6 t ^{235}U, releasing a total energy of 2−3 GWy, at a power level probably $\leq$10 kW. About 1.0−1.5 t of ^{239}Pu was formed by neutron capture in ^{238}U, but the relatively short (on a geological scale) half-life of ^{239}Pu allowed it to decay into ^{235}U. Since a few samples enriched in ^{235}U also have been found, it is believed that in some places in Oklo breeding conditions may temporarily have existed, cf. §22.12.

It has been calculated that many uranium rich ore deposits $2-3 \times 10^9$ y ago must have been supercritical in the presence of moderating water. Therefore natural nuclear chain reactions may have had an important local influence on the early environment of earth.

19.11. REACTOR CONCEPTS

Nuclear reactors are designed for production of heat, mechanical and electric power, radioactive nuclides, weapons material, research in nuclear physics and chemistry, etc. The design depends on the purposes, e.g. in the case of electric power production the design is chosen to provide the cheapest electricity taking long term reliability in consideration. This may be modified by the availability and economy of national resources such as raw material, manpower and skill, safety reasons, etc. Also the risk for proliferation of reactor

[5] n-fluxes and fluencies are given per m^2 in the SI system; however, it is still common to use per cm^2.